

The Four Types of Agent Memory: Working, Semantic, Procedural, Episodic

Know which store to reach for, what each costs you in context, and why forgetting is the hard part.

For engineers building agents who have to decide what the system remembers, and where · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

[Watch the source video](#)[Read the full lesson](#)[Read the condensed version](#)

What matters

- ✓ Models are stateless — every kind of agent memory is machinery the surrounding system provides.
- ✓ CoALA names four types: working, semantic, procedural and episodic, mirroring human memory.
- ✓ Working memory is the context window: fast, always present, volatile, and capped in size.
- ✓ Semantic memory is what the agent knows in general, and in production it is often just Markdown files rather than a vector database.
- ✓ Procedural memory is how the agent does things, packaged as skills with step-by-step instructions.
- ✓ Progressive disclosure keeps procedural memory affordable: advertise a short index, load the full instructions only on a match.
- ✓ Episodic memory is distilled experience, not saved transcripts — the compression is what makes it useful.
- ✓ Forgetting is the hardest part, because staleness and obsolescence have to be decided in code rather than felt.

What it covers

- 01 Human memory as the map** 0:00
Four kinds of remembering that people do without noticing, which turn out to be four separate engineering problems.
- 02 CoALA: naming the four** 1:25
A framework from a Princeton research team that maps out the distinct memory types an agent needs.
- 03 Working memory: the context window** 2:09
Everything the agent can see right now — the conversation, the system instructions, whatever has been loaded into the prompt.
- 04 Semantic memory: what the agent knows** 3:33
The knowledge base — facts, rules, conventions, documentation. In production it is frequently just Markdown.
- 05 Procedural memory: how the agent does things** 4:59
Skills — a folder with a Markdown file describing what the skill does and the steps to perform it.
- 06 Progressive disclosure** 5:41
The agent advertises a lightweight index, loads full instructions on a match, and pulls in supporting files only during execution.
- 07 Episodic memory: what happened before** 6:25
The record of past interactions and decisions, and what was learned from them — distilled rather than stored whole.

08 Forgetting is an engineering problem 7:52

What do you delete? When does something become obsolete? People forget effortlessly; agents have to be told.

09 Not every agent needs all four 8:32

Match the memory architecture to what the agent actually does. Most need fewer types than the taxonomy suggests.

Glossary

CoALA — Cognitive Architectures for Language Agents, a framework from a Princeton research team that identifies four memory types agents need.

Semantic memory — What the agent knows in general — facts, rules, conventions and documentation. Often a Markdown file loaded at session start.

Episodic memory — The record of past sessions and what was learned from them, kept as distilled notes rather than whole transcripts.

Progressive disclosure — Advertising skills by a short description, loading full instructions only on a match, and supporting files only during execution.

SKILL.md — The Markdown file defining a skill: what it does, when to use it, and the steps to perform it.

Distillation — Compressing a session into a few durable conclusions worth remembering, instead of storing the transcript.

Reflex agent — An agent that maps an input directly to an action with no persistence, needing only working memory.

Working memory — The context window: the conversation, instructions and loaded data the agent can see right now. Fast, volatile and capped.

Procedural memory — How the agent performs a task: ordered, step-by-step instructions, packaged as a skill.

Context window — The maximum text a model can process in one call. It bounds working memory and everything loaded into it.

Skill index — The lightweight list of skill names and descriptions the agent always sees — roughly a hundred tokens per skill.

CLAUDE.md — A project-root Markdown file holding architecture, conventions, commands and prohibitions, loaded at the start of every session.

Scoped invalidation — Dropping a whole class of memories when the context they belonged to ends — a project archived, an account closed.

Stale note — A stored memory that has become false. Worse than no memory, because it displaces correct behaviour rather than merely omitting it.

Explanations are AI-generated. Check anything you intend to rely on.